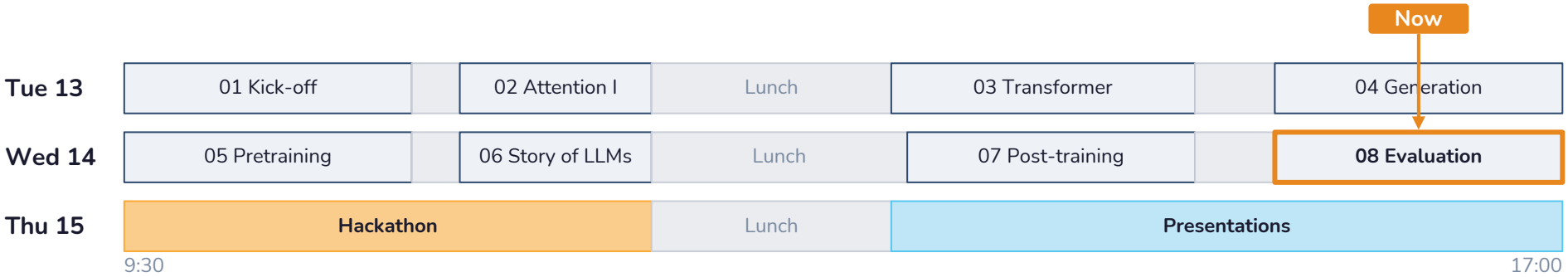
Advanced NLP

Session 08 · Evaluating LLMs

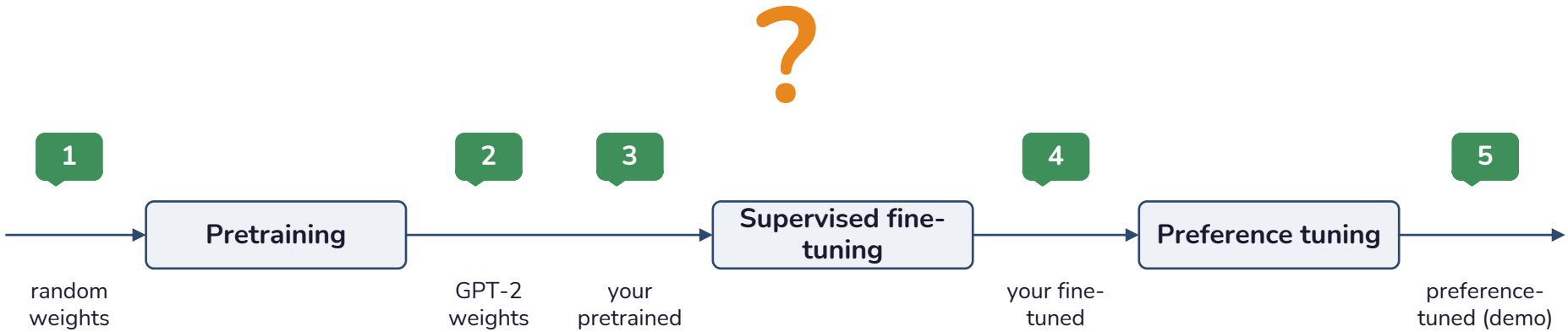
Prof. Dr. Richard Sieg · TH Köln IWS · WS 26 / 27
Wednesday, October 14, 2026

Which model is best, and what that means tomorrow

Time	
15:30	Two metrics, benchmarks, judges
15:59	Lab: notebook 08
16:19	Thursday's project
16:41	Your component
17:00	End of day 2



You have five models now. Which one is best?



A measurement only tells you what it measures, and only above the noise.

Agenda

01. Two metrics

02. Benchmarks

03. Judges

04. Thursday's project

01.

Two metrics

Perplexity says the pretrained model is best

Model	Perplexity on held-out TinyStories
Pretrained	5.12
SFT	5.61
DPO	5.89

- Lower is better. The pretrained model was trained on exactly this kind of text.
- Fine-tuning and preference tuning moved the model away from plain stories, towards following instructions.

The required-words rate says exactly the opposite

Model	Perplexity	Required words used
Þŕétŕáıńéð	5.12, **ƅéśt**	15.3%, **ẃóŕśt**
ŚGT	5.61	44.7%
DÞÓ	5.89, **ẃóŕśt**	70.0%, **ƅéśt**

- Same three models, opposite rankings.
- Perplexity measures how well a model fits a kind of text. The word rate measures whether it does a task.

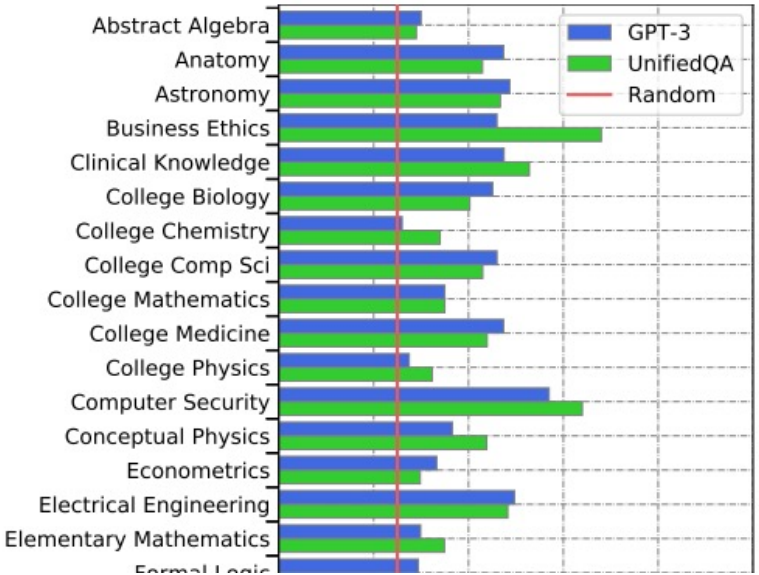
Neither number is “the” score. Each answers its own question.

02.

Benchmarks

MMLU: 57 knowledge tasks, multiple choice

- Massive Multitask Language Understanding
- Contains 57 diverse knowledge tasks
- Was the “go to” LLM benchmark until recently
- Benchmarks can’t keep up with LLM development



Why isn’t there a planet where the asteroid belt is located?
(A) A planet once formed here but it was broken apart by a catastrophic collision.
(B) There was not enough material in this part of the solar nebula to form a planet.
(C) There was too much rocky material to form a terrestrial planet but not enough gaseous material to form a jovian planet.
(D) Resonance with Jupiter prevented material from collecting together to form a planet.

Figure 16: An Astronomy example.

If each of the following meals provides the same number of calories, which meal requires the most land to produce the food?
(A) Red beans and rice
(B) Steak and a baked potato
(C) Corn tortilla and refried beans
(D) Lentil soup and brown bread

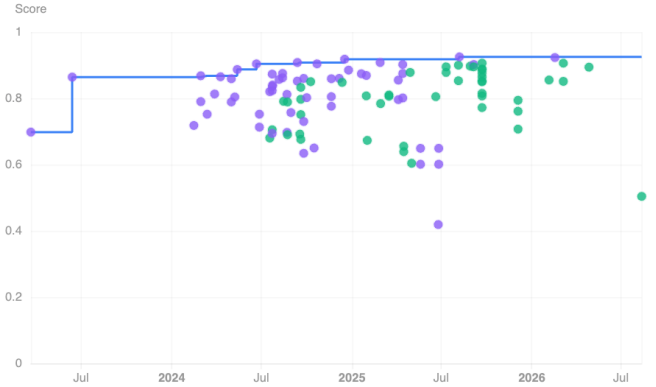
Measuring Massive Multitask Language Understanding [Hendrycks et al, 2020]

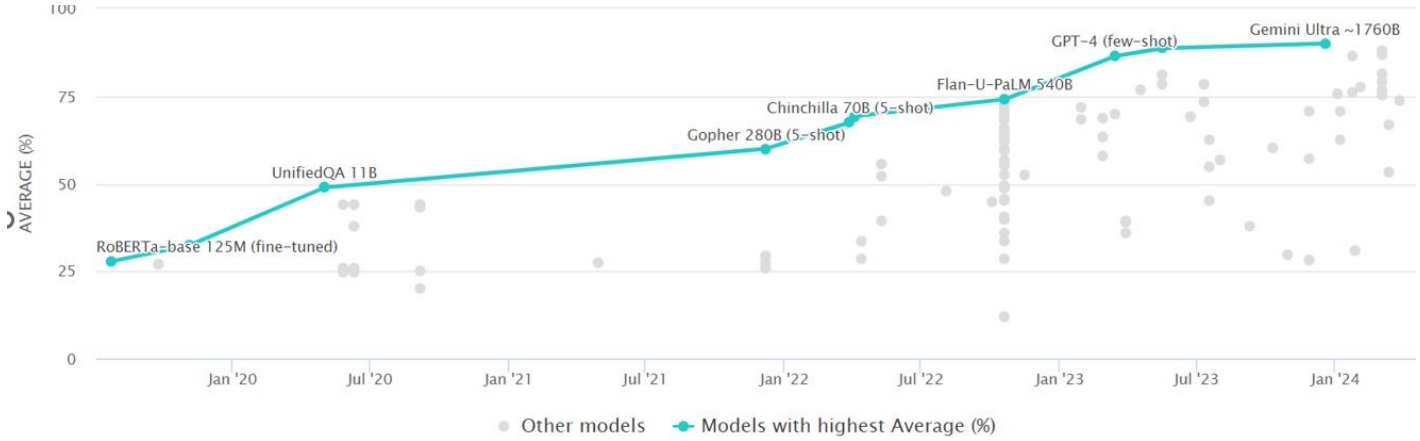
MMLU went from hard to nearly solved

LLM Stats Leaderboard

GPT3 was SOTA

Model	Humanities	Social Science	STEM	Other	Average
Random Baseline	25.0	25.0	25.0	25.0	25.0
RoBERTa	27.9	28.8	27.0	27.7	27.9
ALBERT	27.2	25.7	27.7	27.9	27.1
GPT-2	32.8	33.3	30.2	33.1	32.4
UnifiedQA	45.6	56.6	40.2	54.6	48.9
GPT-3 Small (few-shot)	24.4	30.9	26.0	24.1	25.9
GPT-3 Medium (few-shot)	26.1	21.6	25.6	25.5	24.9
GPT-3 Large (few-shot)	27.1	25.6	24.3	26.5	26.0
GPT-3 X-Large (few-shot)	40.8	50.4	36.7	48.8	43.9

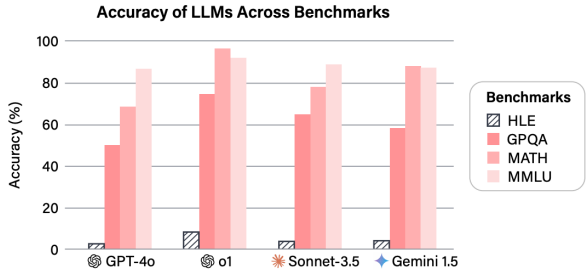


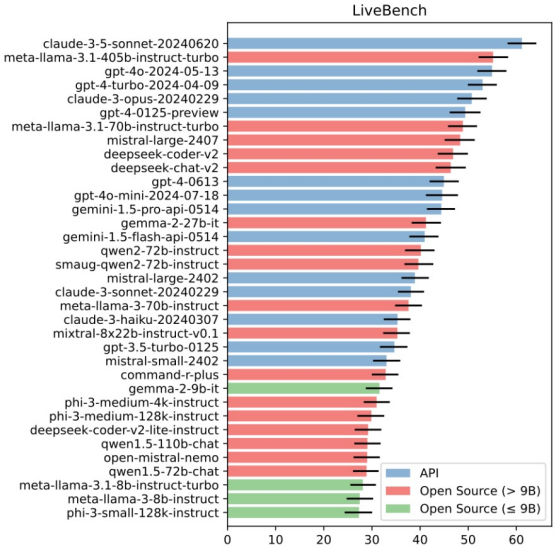


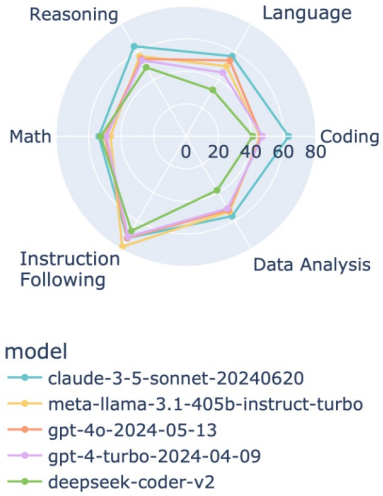
Measuring Massive Multitask Language Understanding [Hendrycks et al, 2020]

New benchmarks keep replacing the solved ones

- BIG-Bench (Google, 2022) contains 204 tasks
- Humanity's Last Exam (AI Safety, 2025)
- LiveBench (2024) gets updated monthly,
  to limit potential contamination into the training data of LLMs







Our model scores 25 percent on benchmarks: chance

- Benchmarks need scale. HellaSwag, ARC and the other standard suites: four answer options, written for models with billions of parameters and broad world knowledge.
- A 17.7 million parameter model that has only read children's stories scores at chance: 25 percent, whichever of the three checkpoints you pick.
- That is why nobody benchmarks a model this size, and why our metrics this week were loss and a task we could check.

Contamination inflates scores, saturation ends them

- ***Contamination:*** the benchmark's questions end up in the training data. The model recognizes answers instead of working them out, like a student who saw the exam.
- ***Saturation:*** every model scores near the top, and the benchmark no longer tells them apart.
- Assume any public dataset is or will be contaminated.
- Defenses: a canary string to search training data for, gated or encrypted test sets, benchmarks that change over time, like LiveBench.

Fourrier et al., The LLM Evaluation Guidebook: huggingface.co/spaces/OpenEvals/evaluation-guidebook

Leaderboard numbers are hard to reproduce

- The same model scores differently with a different prompt template, a different number of examples in the prompt, or different examples.
- With sampling, the seed and the temperature matter. For reasoning models, whether thinking is switched on, and how many tokens it may use.
- Labs choose which checkpoint, which prompt and which run to report.

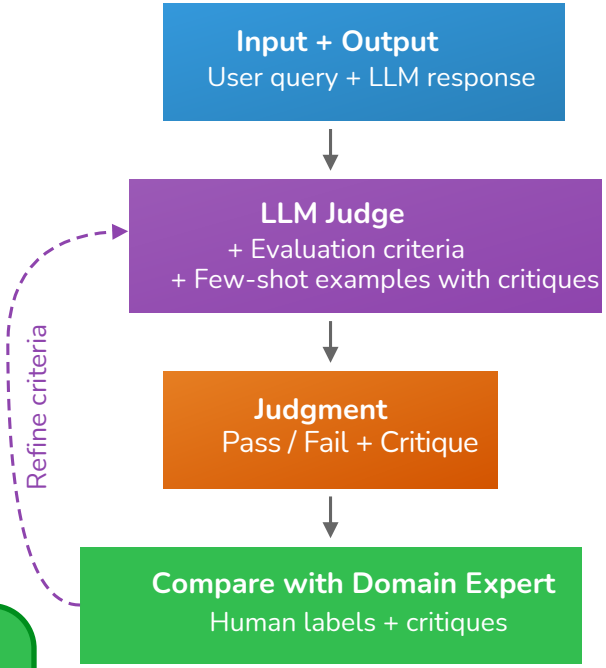
Fourrier et al., The LLM Evaluation Guidebook: huggingface.co/spaces/OpenEvals/evaluation-guidebook

03.

Judges

An LLM can judge outputs when humans cannot keep up

- Issue: Human annotation does not scale
- ***LLM as a Judge***
  - LLM receives input, output, and evaluation criteria
  - LLM produces judgement and explanation (critique)



Common Pitfalls

- Position bias (prefers first answer)
- Length bias (prefers longer answer)
- Self-preference

Best Practices

- Binary pass/fail decision
- Include critique
- Use task-specific criteria

Judges have biases, each with a countermeasure

Bias	What happens	Countermeasure
Þóśıţıóń	prefers the first, or the second, answer	swap the order at random
Véŕƀóśıţý	prefers longer answers	account for length
Śéĺg-þŕégéŕéńćé	prefers outputs like its own	a jury of different judges
Iɲćóńśıśţéńćý	changes its verdict when asked again	ask several times, take the majority
Góŕṃát	judges badly outside its training format	follow the format it was trained on

Zheng et al. 2023, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena: arxiv.org/abs/2306.05685

Every metric this week could be gamed

- ***Perplexity:*** train on the test distribution.
- ***The required-words rate:*** repeat the words. Tom was slow, but he was slow.
- ***A benchmark:*** see its questions during training.
- ***A judge:*** write longer answers, or answer first.
- IFEval checks rules like word counts automatically, over hundreds of prompts: our required-words rule at scale, with the same weakness.

When a measure becomes a target, it stops being a good measure.



Zhou et al. 2023, Instruction-Following Evaluation for LLMs (IFEval): arxiv.org/abs/2311.07911

Tutorial

Notebook 08: measure the three models yourself

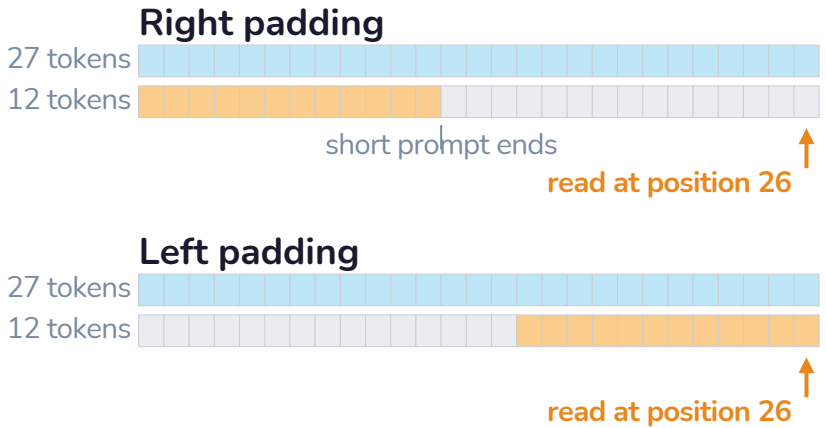
1. Implement `compute_perplexity`: build batches, average the loss, exponentiate.
2. Run the table for the three checkpoints. Do you get the same ranking?
3. Run the right-padding cell, then the left-padding one. Compare what each predicts for the short prompt.
4. Read three completions of the DPO model and three of the SFT model. Which would you rather read?

About 20 minutes, in pairs

- ☐ Your perplexity table matches the slide's ranking
- ☐ You can say why right padding reads the wrong position

The right-padding bug gives a plausible answer

- Two prompts of 27 and 12 tokens, padded on the right to 27.
- The model reads its next token at position 26 for both: 15 positions past the end of the short prompt.
- It predicts “Once” for both. A plausible start of a story, and still the answer to a question nobody asked.
- With left padding, each prompt ends at the last position: the fox story and a story about a boy called Jack, each continuing its own prompt.



A bug that produces plausible output is worse than one that crashes.

04.

Thursday's project

Thursday: does one change help, and can you tell?

GPT-2, and your model	2026 open models	Thursday
LayerNorm	RMSNorm	01
Pre-LN	pre-LN, with post-LN variants back in OLMo 2	01
GELU feed-forward	SwiGLU, often as a mixture of experts	02
Full causal attention	sliding window in some layers	03
Raw query-key scores	QK-Norm	04
Learned positions	RoPE	05, 06
Multi-head attention	grouped-query attention	07
Generation without a cache	KV cache	08
Dropout 0.1	no dropout	

Your folder: task, code, tests and harness

- `TASK.md`: what the component is, what it replaces, exactly what to implement, which runs and which measurements.
- `starter.py`: the classes and functions, named, with their arguments. The bodies are yours.
- `test_shapes.py`: run it first. It tells you the wiring is right, not that the component is.
- `test_sanity.py`: limit cases, for example “with the window as long as the context, sliding-window attention equals normal attention”.
- `run_ablation.py` (one level up, shared): the runner. A quick check of 200 steps in about a minute, then your three seeds under the fixed protocol, with loss curves and the effect against the noise floor.
- `ablation_harness.py` and `baseline.py` (one level up, shared): the fixed protocol and the baseline. Do not edit.
- A short reading list: the original paper first, an overview post afterwards.

```
components/
├─ 03_sliding_window/
│   ├─ TASK.md
│   ├─ starter.py
│   ├─ test_shapes.py
│   └─ test_sanity.py
├─ run_ablation.py
├─ ablation_harness.py
├─ baseline.py
└─ README.md
```

one of eight component folders, 01 to 08

Change one component, and nothing else

- ***The baseline:*** your GPT-2, the small configuration, unchanged, 2,000 steps, seeds 0, 1 and 2.
- ***Your component:*** the same 2,000 steps, the same seeds, the same data, learning rate, batch and context.
- ***One change*** means the component and everything it needs: RoPE also removes the learned positions. It is still one change.
- ***Parameter count:*** some components change it. Report it, do not compensate for it, discuss it.

Validation loss is the right metric here

A metric for training experiments should, according to the team behind FineWeb:

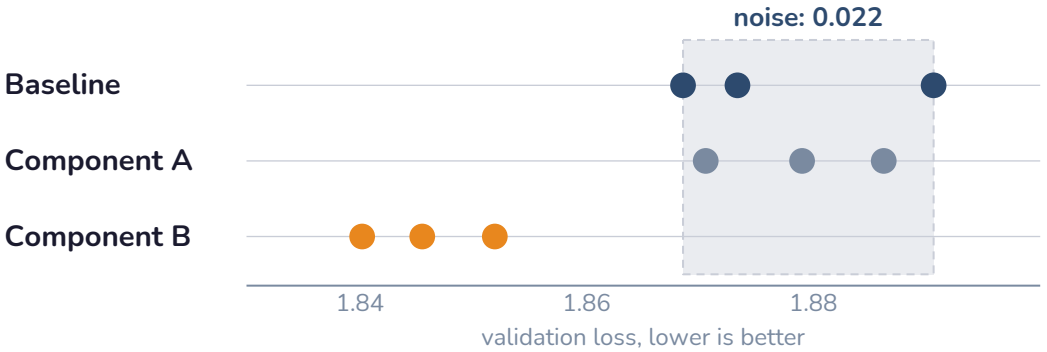
- ✓ score above chance
- ✓ improve as training goes on
- ✓ vary little between seeds
- ✓ rank models consistently

Validation loss on our fixed validation set does all four. Benchmarks, at our size, do none.

Fourrier et al., The LLM Evaluation Guidebook: huggingface.co/spaces/OpenEvals/evaluation-guidebook

Smaller than 0.022 is not an effect

Seed	Baseline validation loss
0	1.8733
1	1.8906
2	1.8685
Spread	0.022



A: inside the noise, inconclusive. B: outside it, an effect. Points for A and B are illustrative.

- Training is seeded but not bit-for-bit identical across machines. Your three baseline runs should land within the noise floor of these numbers. For example, on an A100 the same setup gave 1.8657, 1.8911 and 1.8734.
- If your component moves the mean by less than 0.022, the honest answer is: inconclusive at this scale. That scores as well as a clear win.

An effect inside the noise is no effect.

Each component has a second metric

for what it was built to improve

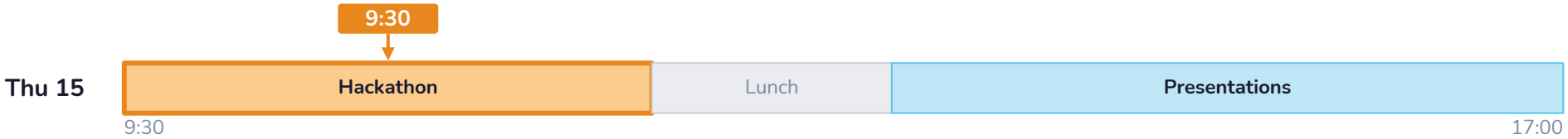
Component	Second metric
RMSNorm, norm placement	step time; training stability
SwiGLU	loss at matched parameters
Sliding window	loss by position, and window size
QK-Norm	largest attention logit at the end of training
Sinusoidal positions	loss by position
RoPE	loss by position
Grouped-query attention	KV cache size
KV cache	tokens per second, on GPU and CPU

Thursday is graded on the experiment, not its result

- ***Presentation, Thursday afternoon, 60 percent:*** 12 minutes per pair, then 6 minutes of questions, to one of you at a time, both of you asked.
- ***Report and repository, October 21, 40 percent:*** four pages, with a section on what did not work, and the appendix on AI use.
- Whether your component helps is not graded. Whether you can tell is.

Thursday, hour by hour

Time	What
9:30	Start your baseline with the runner, three seeds, about 20 minutes, on your second GPU. Then a short briefing
9:45	Implement your component, run its two tests, then the quick check (200 steps, about a minute)
from 10:30	Component runs, same three seeds, 5 to 7 minutes per seed. A checkpoint with each pair
12:15	Lunch, runs can keep going
13:30	Presentations, 18 minutes per pair



A measurement only tells you what it measures, and only above the noise.

Your component

Pair	Component
1	
2	
3	
4	
5	
6	
7	
8	

Tonight: read your task, and one chapter

- Your component's `TASK.md`, and the first paper on its reading list.
- The ablation chapter of the Smol Training Playbook: how a team that trains real models runs exactly this kind of experiment.
- Run your shape test once. It should fail with “not implemented”, not with an import error.

Sources

Literature

- Fourrier et al., The LLM Evaluation Guidebook, Hugging Face: contamination and saturation, reproducing published scores, judge biases and their mitigations, choosing evaluations for training ablations.
- Hendrycks et al. 2020, Measuring Massive Multitask Language Understanding, arxiv.org/abs/2009.03300.
- Zheng et al. 2023, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena, arxiv.org/abs/2306.05685. Position bias.

Literature, continued

- Zhou et al. 2023, Instruction-Following Evaluation for Large Language Models, arxiv.org/abs/2311.07911. IFEval.
- Eldan, Li 2023, TinyStories, arxiv.org/abs/2305.07759. Grading by GPT-4.
- The Smol Training Playbook, Hugging Face. The ablation chapter.

Not presented, included in the handout.